Operational Anomaly & Cost Impact Analysis
Amazon-Inspired Supply Chain Operations Project

Project Report Submitted by	Pranjal Gupta
Domain	Supply Chain Analytics / Operations Analytics / Business Analysis
Tools & Technologies Used	Excel | Python | Machine Learning | SQL
Techniques Applied	Data Engineering • Exploratory Data Analysis (EDA) • Geospatial Analytics • Business Analytics • SQL techniques • • Machine Learning •
Project Difficulty Level	Intermediate

Dataset Availability

The cleaned dataset used for this project is publicly available in the GitHub repository:

https://github.com/pranjalgupta4482-ui/amazon-operational-anomaly-analysis

About Dataset

The dataset represents order-level supply chain operations similar to large-scale e-commerce environments.
It includes information related to delivery timelines, operational processes, financial impact, and geographic distribution.

The dataset was used to simulate Amazon-inspired supply chain operations.

Dataset Structure

The project utilized three logical stages of data:

1. Raw Dataset
    - Contained full operational data
    - Included latitude and longitude
    - Had inconsistent column naming and formats
2. Cleaned Dataset
    - Standardized column names
    - Corrected date formats
    - Removed irrelevant attributes
    - Prepared for SQL and Python analysis
3. Enriched Dataset
    - Cleaned data merged with geospatial attributes
    - Final analytical dataset used for EDA, mapping, and ML

This layered approach follows real-world data engineering practices.

Data Quality Challenges

During analysis, the following real-world issues were encountered and resolved:

- Inconsistent date formats
- Character encoding issues
- Column naming mismatches between raw and cleaned files
- Missing values in operational and location data

➢ EXECUTIVE SUMMARY

This project focuses on analysing operational delivery data to identify the key drivers of high-cost impact and to detect anomalous operational events that require immediate attention. Using an Amazon-inspired supply chain dataset, the analysis integrates SQL-based business analysis, Python exploratory data analysis, geospatial visualization, and machine learning to provide actionable insights.

- ➢ Implemented an unsupervised machine learning model (Isolation Forest) to address the absence of labelled risk outcomes.
- ➢ Automatically identified high-risk operational anomalies based on unusual combinations of:

- Delivery delay duration
- Cost impact
- Process severity

- ➢ Flagged anomalous events as priority cases for proactive operational intervention.
- ➢ Demonstrated that early detection of long delays, particularly within complex process types, can significantly reduce operational losses.
- ➢ Showed how data analytics and machine learning enable improved cost control, operational efficiency, and informed decision-making in modern supply chain environments.

BUSINESS PROBLEM STATEMENT

Large-scale supply chain operations experience frequent delivery delays; however, only a small subset of these delays results in significant operational cost and business risk. Existing reporting approaches lack the ability to identify high-impact delays early, leading to reactive interventions and avoidable cost escalation.

The challenge lies in prioritizing high-risk operational events in the absence of labelled outcomes, while accounting for variations in process complexity and geographic distribution.

PROJECT OBJECTIVE

The objective of this project is to identify and prioritize high-risk delivery delays that disproportionately increase operational costs by applying data analytics and machine learning techniques, enabling early detection, focused monitoring, and proactive operational decision-making.

High-quality analysis depends on clean and consistent data. The raw supply chain dataset contained several real-world data quality issues that required systematic cleaning and preparation before analysis.

Key Data Cleaning Steps Performed:

- **Date Standardization:**
  Delivery and shipping date fields were recorded in inconsistent formats. All date columns were standardized to a uniform datetime format to enable accurate delay calculations and time-based analysis.
- **Handling Encoding Issues:**
  The dataset contained non-UTF-8 characters originating from Excel exports. Appropriate encoding methods were applied during data ingestion to prevent data loss and parsing errors.
- **Column Selection and Renaming:**
  Irrelevant attributes were removed, and column names were standardized to maintain consistency across raw, cleaned, and analytical datasets.
- **Missing Value Handling:**
  Missing values in operational and financial fields were identified and handled using appropriate techniques to ensure analytical and modelling stability.

- **Geospatial Data Integration:**
  Latitude and longitude information available only in the raw dataset were extracted and merged with the cleaned dataset using a common event identifier, resulting in a single enriched dataset suitable for geospatial analysis and machine learning.
- **Data Validation:**
  post-cleaning validation checks were performed to confirm data integrity, correct joins, and logical consistency across key variables.

SQL ANALYSIS

Structured SQL analysis was performed to examine delivery performance, cost behaviour, and operational patterns at scale.

Purpose of SQL Analysis

The SQL layer was used to:

- Establish baseline operational performance
- Identify delay and cost patterns across processes and regions
- Narrow down high-risk operational events for deeper analysis in Python

Key SQL Analyses Conducted

- **On-Time vs Delayed Orders:**
  Evaluated the proportion of delayed versus on-time orders to understand overall delivery performance and delay prevalence.
- **Delay Analysis by Process Type:**
  Calculated average delay days across different process types to identify workflows with consistently higher operational delays.
- **Cost Impact Assessment:**
  Compared cost impact between delayed and on-time orders to quantify the financial consequences of delivery delays.

- **High-Risk Event Identification:**
  Filtered and ranked orders with extreme delay durations and high cost impact to highlight potential operational risk cases.
- **Temporal Trend Analysis:**
  Analysed monthly delay trends to detect seasonal or time-based patterns affecting delivery performance.

Key Insights from SQL Analysis

- The majority of orders were delivered on time or with minimal delay.
- A relatively small subset of delayed orders accounted for a disproportionate share of operational cost.
- Certain process types consistently exhibited higher average delays, indicating structural inefficiencies.
- Delay patterns varied over time, suggesting the influence of operational load and seasonality.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted using Python to uncover patterns, relationships, and operational behaviours that are not easily captured through aggregate SQL queries alone. The objective of EDA was to understand the drivers of delivery delays and cost impact, and to guide feature selection for machine learning.

1. Delivery Delay Distribution

The distribution of delivery delay days was analyzed to understand the overall nature of delays across the dataset.

Observations:

- Most orders experienced **no delay or minimal delay**.
- A small number of orders showed **significantly higher delay durations**.

Insight:

Delivery delays are **skewed rather than evenly distributed**, indicating that extreme delays are rare but operationally significant.

2. Relationship Between Delay and Cost Impact

The relationship between delivery delay days and cost impact was examined using statistical and visual analysis.

Observations:

- Cost impact remains relatively stable for short delays.
- Cost increases sharply as delay duration increases beyond certain thresholds.

Insight:

Operational cost increases **non-linearly** with delivery delay, highlighting that long delays are disproportionately expensive.

3. Process Type Analysis

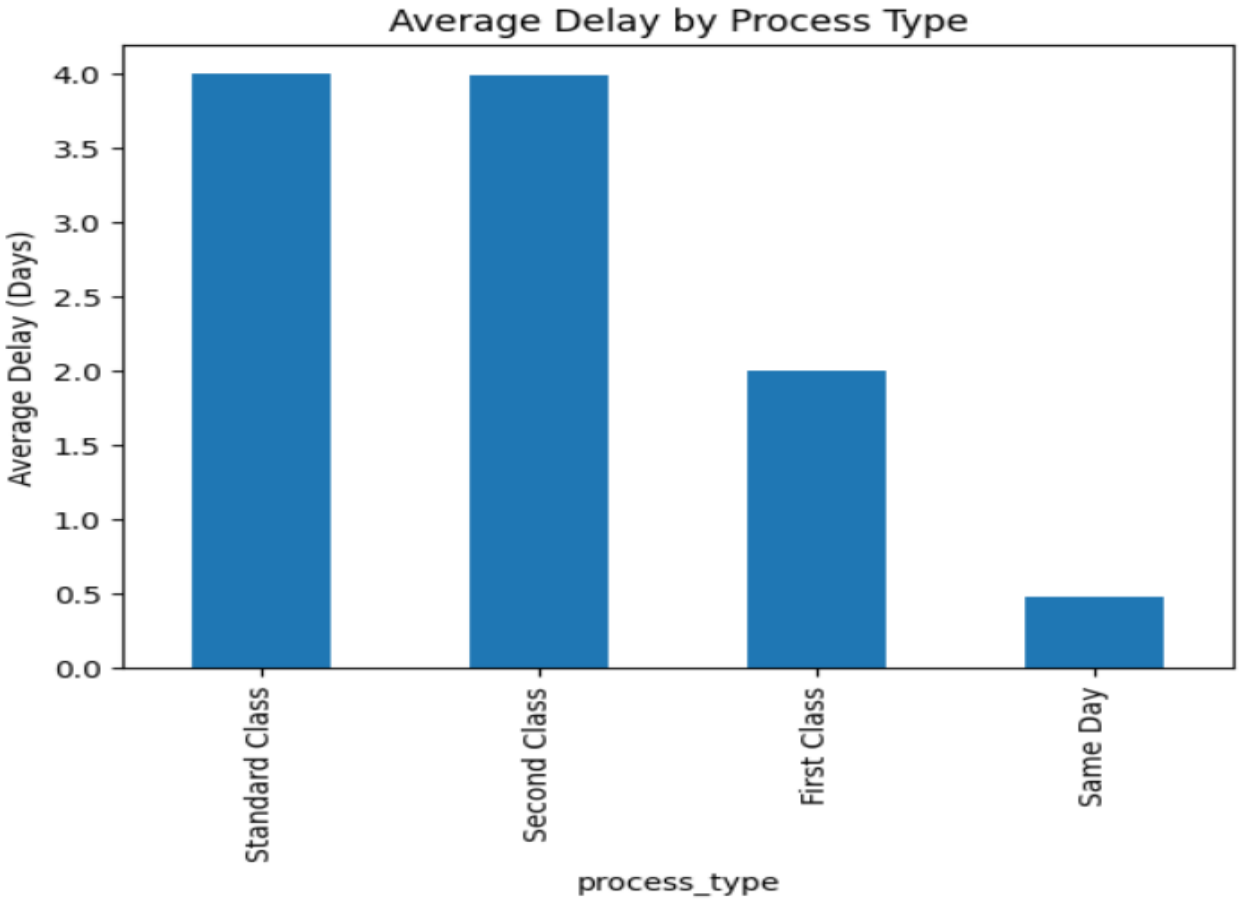
EDA was used to compare delay and cost behaviour across different process types.

Observations:

- Certain process types consistently showed higher average delays.
- The same delay duration resulted in different cost impacts depending on the process type.

Insight:

Process complexity amplifies cost impact, making some workflows more vulnerable to delays than others.



4. Regional and Geographic Patterns

Operational performance was analysed across regions to identify geographic variation in delivery behaviour.

Observations:

- Delay frequency and severity varied across regions.
- Some regions showed consistent clustering of delayed orders.

Insight:

Geographic factors contribute to delivery performance, indicating region-specific operational constraints or capacity limitations.

GEOSPATIAL ANALYSIS

Geospatial analysis was conducted to incorporate location-based context into the evaluation of delivery delays and operational cost impact. By mapping delivery events geographically, the analysis aimed to identify regional delay hotspots and uncover spatial patterns that are not visible through tabular analysis alone.

Approach

- Latitude and longitude information was extracted from the raw dataset and merged into the cleaned analytical dataset using a common event identifier.
- An interactive map was created using Python and Folium.
- Delivery locations were plotted as point markers and color-coded based on delay severity.

Key Observations

- Delayed deliveries were not uniformly distributed across locations.
- Certain regions exhibited higher concentrations of delayed and high-cost orders, forming identifiable delay hotspots.

- High-cost operational anomalies tended to cluster geographically rather than occurring randomly.

Business Value

Geospatial analysis provides operations teams with:

- Improved visibility into where delays and high-cost events are occurring.
- The ability to design region-specific operational strategies.
- Support for proactive planning, such as rerouting, capacity enhancement, or localized process optimization.

MACHINE LEARNING

Machine learning was applied to automatically identify high-risk operational events that contribute disproportionately to operational cost. Given the absence of labelled outcomes in the dataset, an unsupervised learning approach was selected to detect anomalous patterns in delivery and cost behaviour

Rationale for Using Isolation Forest:

- Does not require labelled target variables
- Efficiently isolates rare, high-impact events
- Widely used in operational risk and fraud detection scenarios

Business Relevance

The machine learning approach enables:

- Early identification of high-risk operational events
- Proactive decision-making instead of reactive handling
- Improved focus on cost-intensive operational failures

By integrating machine learning into the analytics workflow, the project demonstrates how advanced analytics can support operational efficiency and cost control in modern supply chain environments.

KEY FINDINGS

- Most delivery operations are completed on time with minimal cost impact.
- A small subset of delayed orders accounts for a disproportionate share of operational cost.
- Operational cost increases non-linearly as delivery delays extend.
- Process type significantly influences cost escalation, with complex workflows amplifying delay impact.
- High-cost operational events tend to cluster geographically.
- Machine learning successfully identified high-risk anomalies that are not evident through averages alone.

BUSINESS IMPACT

- Enables operations teams to focus on high-impact delays rather than all delayed orders.
- Supports early identification of cost-intensive operational risks.
- Improves resource prioritization and escalation efficiency.
- Enhances decision-making through data-driven and location-aware insights.
- Reduces avoidable operational losses through proactive intervention.

RECOMMENDATIONS

- Implement early-warning systems to detect long delays before cost escalates.
- Closely monitor high-risk process types with greater operational complexity.
- Use geospatial insights to optimize regional operations and capacity planning.

- Integrate anomaly detection outputs into regular operational reviews.

CONCLUSION

- This project demonstrates how an end-to-end data analytics approach can be used to identify and manage high-risk operational events in large-scale supply chain environments. By integrating SQL analysis, Python-based exploratory analysis, geospatial visualization, and unsupervised machine learning, the project successfully identified the key drivers of operational cost escalation.
- The analysis shows that long delivery delays within complex process types are the primary contributors to high operational costs, and that these events can be effectively detected using anomaly-based machine learning techniques. The findings highlight the value of early detection, targeted monitoring, and data-driven decision-making in reducing operational losses.
- Overall, the project illustrates how data analytics and machine learning can support cost control, operational efficiency, and proactive risk management, making it a practical and scalable solution for modern supply chain operations.